

Tutorial 15 Currents Suggested Solutions

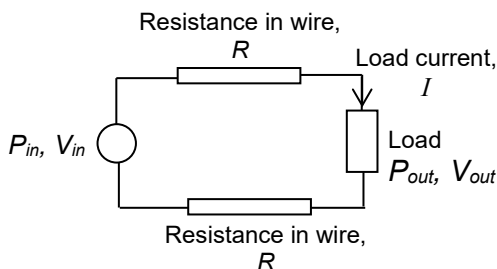
- D1** (a) $I_n = (1.56 \times 10^{15} \text{ s}^{-1})(1.60 \times 10^{-19} \text{ C}) = 0.250 \times 10^{-3} \text{ C s}^{-1}$ [1]
- (b) $I = I_n + I_p$
 $I_p = I - I_n = 0.28 - 0.250 = 0.03 \text{ mA}$ (2 d.p. – follow least d.p.) [1]
- (c) No. of positive charges = $(0.03 \times 10^{-3}) / (3.20 \times 10^{-19}) = 9 \times 10^{13} \text{ s}^{-1}$ (1 s.f.) [1]
- (d) This is because current 0.28 mA (2 d.p.) is not precise, leading to a poor precision in the positive current (only 1 s.f.) and hence the number of positive charges per unit time. [1]

D2 Number density $n = \frac{\rho_{\text{copper}}}{m_{\text{copper atom}}} = \frac{8960}{1.06 \times 10^{-25}} = 8.45 \times 10^{28} \text{ m}^{-3}$. [1]

$$I = nAvq$$

$$\Rightarrow v = \frac{I}{nAq} = \frac{1.0}{8.45 \times 10^{28} \times \pi \times (1.0 \times 10^{-3} / 2)^2 \times 1.60 \times 10^{-19}} = 0.094 \text{ mm s}^{-1}$$
 [1]

- D3** (a) One copper wire carries current to the load, the other carries current from the load to form a closed circuit.



$$R = 50.0\text{m} \times \frac{0.108\Omega}{300\text{m}} = 0.018 \Omega$$
 [1]

$$V_{in} = 2(RI) + V_{out}$$

$$V_{out} = V_{in} - 2(RI) = 120 - 2(0.018 \times 110) = 120 - 3.96 = 116.04 = 116 \text{ V}$$
 [1]

(b) (i) $P_{out} = IV_{out} = 110(116.04) = 12\,764.4 = 12\,800 \text{ W} = 12.8 \text{ kW}$ [1]

(ii) $P_{in} = P_{lost} + P_{out}$ where P_{lost} is the power lost in the 2 copper wires

$$P_{lost} = IV_{in} - P_{out} = 110(120) - 12\,764.4 = 435.6$$
 [1]

$$= 436 \text{ W}$$
 [1]

Different methods of getting P_{lost} include:

$$P_{lost} = 2(I^2 R) = 2(110)^2(0.018) = 435.6 = 436 \text{ W}$$

OR

$$P_{lost} = IV_{wire} = (110)(3.96) = 435.6 = 436 \text{ W}$$

OR

$$P_{lost} = \frac{V_{wire}^2}{R_{wire}} = \frac{(3.96)^2}{2 \times 0.018} = 435.6 = 436 \text{ W}$$

(iii) % efficiency = $\frac{P_{out}}{P_{in}} \times 100\% = \frac{12800}{(110)(120)} \times 100 = 97\%$ [1]

(c) Amount customer needs to pay = $P_{out} \times 3 \times 0.17$
 $= 12.8 \times 3 \times 0.17$
 $= \$6.53$ [1]

D4 (a) $I = \frac{P_{supply}}{V_{supply}} = \frac{6700 \times 10^3}{25000} = 268 \text{ A}$ [1]

(b) (i) $R = 0.344 \times 30.0 = 10.32 = 10.3 \Omega$ [1]

(ii) $V_{train} = V_{supply} - V_{wire} = 25000 - 241 \times 10.32 = 22512.9$ [1]
 $= 22500 \text{ V}$ [1]

(iii) $P_{train} = I V_{train} = 241 \times 22512.9 = 5425.6 \text{ kW} = 5430 \text{ kW}$ [1]

(iv) $f = \frac{P_{train}}{P_{supply}} = \frac{5425600}{241 \times 25000} = 0.900$ [1]

- (c) (i) • Railway employee and track are both grounded, i.e., they have the same potential and hence there is no p.d. between hand and feet of employee. [1]
 • Therefore, no current flows between the track and employee. [1]
- (ii) • Power transmitted is $P = IV$.
 • High voltage supply $\Rightarrow$ low current. [1]
 • Power loss due to joule heating (I^2R) is minimised. [1]
- (iii) • Additional power is required as train gains G.P.E. while climbing. [1]
 • Assuming same voltage is supplied, higher current is required. [1]

D5 Ans: C

In order to melt identical fuse, power dissipated in the fuse by the d.c. source must be equal to the mean power dissipated in the fuse by the a.c. source:

$$P_{dc} = \langle P_{ac} \rangle$$

$$I_{dc}^2 R = \langle I_{ac}^2 \rangle R$$

$$I_{dc} = \sqrt{\langle I_{ac}^2 \rangle}$$

$$= I_{rms}$$

Hence, if a fuse melts when alternating current just exceeds 13 A r.m.s., it will also melt when direct current just exceeds 13 A.

D6 In Fig. 2a, mean power $P = \frac{V_{r.m.s.1}^2}{R} = \frac{(V_0/\sqrt{2})^2}{R} = \frac{V_0^2}{2R}$ [1]

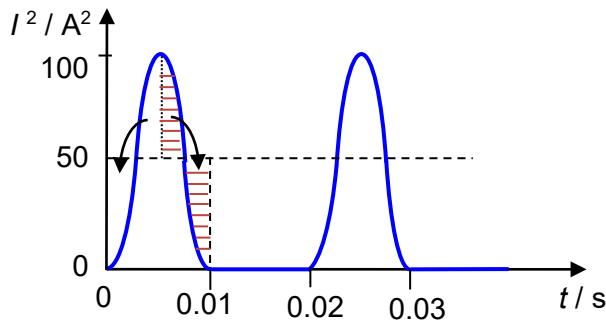
In Fig. 2b, mean power $P' = \frac{V_{r.m.s.2}^2}{R} = \frac{V_0^2}{R} = 2P$ [1]

D7 (a) Case (i) and (ii):

$$I_{\text{r.m.s.}} = \frac{10}{\sqrt{2}} = 7.07 \text{ A} \quad [2]$$

(b) The r.m.s. current of sinusoidal waveforms is independent of frequency or period. [1]

(c) Case (iii): Using graphical method,



1. Square the instantaneous current I

2. Mean square current

= (total area in one period) / (time for one period)

$$= \frac{(50 \times 0.01) + 0}{0.02} = 25 \text{ A}^2$$

3. Take square root of the mean square current = 5 A

Therefore $I_{\text{r.m.s.}} = 5 \text{ A}$

These steps prove that, for half-wave rectified current with peak value I_0 , the r.m.s. current is given by

$$I_{\text{r.m.s.}} = \frac{I_0}{2}$$

[1]

[1]

Alternatively:

For full sinusoidal waveform, mean power delivered is

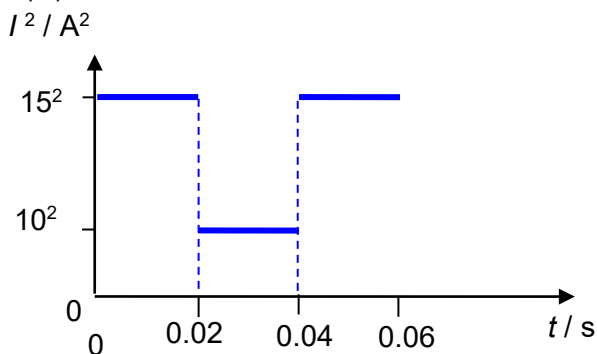
$$\langle P \rangle \propto (I_{\text{r.m.s.}})^2 \quad \text{----- (1)}$$

For half-wave rectified waveform, half the mean power is "lost",

$$\frac{\langle P \rangle}{2} \propto (I'_{\text{r.m.s.}})^2 \quad \text{----- (2)}$$

$$\frac{(2)}{(1)} \Rightarrow \frac{1}{2} = \frac{(I'_{\text{r.m.s.}})^2}{(I_{\text{r.m.s.}})^2} \Rightarrow I'_{\text{r.m.s.}} = \frac{1}{\sqrt{2}} I_{\text{r.m.s.}} = \frac{1}{\sqrt{2}} \left(\frac{I_0}{\sqrt{2}} \right) = \frac{I_0}{2} = \frac{10 \text{ A}}{2} = 5 \text{ A}$$

Case (iv):



$$I_{\text{r.m.s.}} = \sqrt{\frac{(15^2 \times 0.02) + (10^2 \times 0.02)}{0.04}} = 12.7 \text{ A}$$

[2]

- D8** (a) Total time in 2 periods = $5 \text{ cm} \times 0.5 \text{ ms cm}^{-1} = 2.5 \text{ ms}$
Period = 1.25 ms [1]
- (b) Frequency = $1/(1.25 \text{ ms}) = 800 \text{ Hz}$ [1]
- (b) Peak value of potential difference = $(4 \frac{2}{3} \text{ cm} \times 2 \text{ V cm}^{-1}) / 2 = 4.7 \text{ V}$ [1]
- (b) $\frac{4.7}{\sqrt{2}} = 3.3 \text{ V}$
Root-mean-square value of potential difference = [1]

- C1** (a) Power loss in the line, $P_{\text{loss}} = I^2 X = \left(\frac{P}{V}\right)^2 X = \frac{P^2 X}{V^2}$
- $P_{\text{loss}} \propto \frac{1}{V^2}$ since P and X are assumed to be constant.
- (b) $\frac{P_{\text{loss}}}{P_{\text{in}}} \times 100\% \leq 0.015\%$
- $\frac{P^2 X}{V^2} \times \frac{1}{P} \times 100 \leq 0.015$
- $V \geq \sqrt{PX \left(\frac{100}{0.015}\right)}$
- $V \geq \sqrt{(200 \times 10^3)(0.60) \left(\frac{100}{0.015}\right)} \Rightarrow V \geq 2.83 \times 10^4 \text{ V}$
- Hence, minimum output voltage is **$2.83 \times 10^4 \text{ V}$** .

- C2** $I = 1500t - 15$
- $$I_{\text{r.m.s.}} = \sqrt{\frac{\int_0^T I^2 dt}{T}} = \sqrt{\frac{\int_0^T (1500t - 15)^2 dt}{0.02}}$$
- $$= \sqrt{\frac{\int_0^{0.02} (2250000t^2 - 45000t + 225) dt}{0.02}}$$
- $$= \sqrt{\frac{\left[\frac{2250000t^3}{3} - \frac{45000t^2}{2} + 225t \right]_0^{0.02}}{0.02}}$$
- $$= \sqrt{75}$$
- $$= 8.66 \text{ A}$$